
When the Hierarchy Breaks: Failure Motifs and Calibrated Uncertainty from the Many-Body Decomposition of HIP-NN

Working note (anonymous preprint)

Abstract

The Hierarchically Interacting Particle Neural Network (HIP-NN) predicts molecular energies as a sum of per-atom, per-order contributions $\hat{E}_i^n$ and ships with a built-in reliability signal: when the hierarchical terms fail to decay with order n — quantified by the non-hierarchicality R — predictions tend to be worse. The original paper demonstrated this correspondence but left it unexploited: it did not identify *which chemistry* breaks the hierarchy, did not test whether R supports a *calibrated* uncertainty estimate, and did not ask whether hierarchy failures concentrate in a few interpretable structural classes or dilute across chemical space. We answer all three questions in a controlled extrapolation setting: HIP-NN, held fixed at its published architecture and training protocol, is trained on the 20 799 QM9 molecules with at most 8 heavy atoms and interrogated on the 109 024 molecules with exactly 9, where its error tail inflates from 2–5% to 17.9% of molecules above 1 kcal/mol. Against a pre-registered, FDR-controlled library of 30 structural motifs we find that hierarchy failure is chemically structured: 20 motifs form a validated *failure atlas*, led by charge-separated species, cumulated π systems, N=N/N=O units, and polycyclic cage topologies, and for the electronic motifs the atom-resolved non-hierarchicality concentrates up to $6.6\times$ on the motif atoms themselves. A logistic combination of four hierarchy-derived signals from a *single* model gives a calibrated failure predictor (ECE 0.004, AUROC 0.794 on a held-out evaluation half), retains most of the error-filtering utility of a 3-seed deep ensemble at one third of the training cost, and adds discrimination even on top of that ensemble. The failure tail is compact: three motifs cover 80% of all > 1 kcal/mol errors. On 82 288 off-equilibrium ANI-1x conformations the atlas splits cleanly in two: ring-strain topologies transfer as conformational failure motifs (Cliff’s δ up to 0.68), whereas the electronically exotic motifs — and the calibration of R itself — prove specific to near-equilibrium geometry. All conclusions rest on a faithful reimplementation: under the paper’s own random-split protocol at $N_{\text{train+valid}}=50\,000$ our runs reach 0.355 ± 0.003 kcal/mol test MAE against the published 0.354 ± 0.004 , with matching parameter count and energy scale.

1 Introduction

Machine-learning interatomic potentials (MLIPs) for organic molecules have crossed the symbolic “chemical accuracy” threshold of 1 kcal/mol on benchmark datasets: on QM9 [Ramakrishnan et al., 2014], architectures such as HIP-NN [Lubbers et al., 2018], SchNet [Schütt et al., 2018], PhysNet [Unke and Muwly, 2019], and message-passing networks [Gilmer et al., 2017] report mean absolute errors (MAE) of a few tenths of a kcal/mol. The mean, however, is not the quantity that determines whether a chemist can trust a model. Lubbers et al. [2018] report that even at their headline MAE of 0.256 kcal/mol, 2.26% of QM9 test molecules are mispredicted by more than 1 kcal/mol (their Table II); for high-throughput screening, transition-state characterisation, or molecu-

lar dynamics, this one-in-fifty tail — not the mean — is the operative failure rate, and it is invisible unless the model can report *which* predictions to distrust.

HIP-NN is unusually well positioned for this question because its architecture contains a diagnostic that most MLIPs lack. Inspired by the many-body expansion, HIP-NN decomposes the molecular energy as $\hat{E} = \sum_i \sum_n \hat{E}_i^n$, where order n carries roughly $(n+1)$ -body information, and its training loss regularises the terms to decay with n . The original paper defines the non-hierarchicality R (their Eq. 18) and shows in its Fig. 3 that conditional error quantiles rise with R : “if the energy hierarchy produced by HIP-NN does not decay sufficiently fast, the corresponding molecular energy prediction is less likely to be accurate.” That is where the paper stops. The correspondence is left as a correlation on in-distribution test data; nothing in the paper says *what chemistry* makes the hierarchy fail, whether the signal is *calibrated* (does predicted uncertainty match realised error?), or whether failures are *compact* (a few motifs explaining most of the tail) rather than diffuse.

This paper is a direct, quantitative follow-up to that uncertainty claim. We deliberately hold the model fixed — the published architecture, hyperparameters, loss, and training schedule, run in the original authors’ hippynn package [Lubbers et al., 2024] — so that everything we measure is a property of the 2018 model’s hierarchical decomposition, not of a newer architecture. Our design puts the model in a controlled extrapolation setting that mimics how such models are used in practice: we train on the QM9 molecules with ≤ 8 heavy atoms (20 799 training molecules after the paper’s pruning) and analyse behaviour on the 109 024 molecules with exactly 9 heavy atoms — larger, combinatorially richer chemistry than anything seen in training. In this regime the model’s MAE degrades from 0.304 ± 0.010 kcal/mol (in-distribution validation) to 0.781 ± 0.012 kcal/mol, and the fraction of > 1 kcal/mol failures inflates to 17.9% — a tail fat enough to characterise with statistical power, produced by exactly the mechanism (composition extrapolation) that makes uncertainty estimates operationally necessary.

Concretely, we ask and answer three questions.

(Q1) Which chemical motifs break the hierarchy? We pre-registered a library of 30 structural motifs — ring topology and strain, electronic delocalisation, heteroatom–heteroatom bonding, composition, and geometry-derived features — each with an a-priori mechanistic hypothesis connecting it to the decomposition (Sec. 3.2). Testing the full family under Benjamini–Hochberg false-discovery-rate control on a discovery half of the test set, and characterising on a held-out half, yields a *failure atlas* of 20 motifs (Sec. 4.3). The motifs are not merely correlated with error: for the electronically unusual entries the atom-resolved non-hierarchicality R_i concentrates on the motif atoms themselves (up to $6.6\times$), and the per-order energy magnitudes show precisely the failure-to-decay signature the hierarchy hypothesis predicts (Sec. 5).

(Q2) Can the hierarchy’s self-diagnosis be calibrated? We combine four hierarchy-derived signals — R itself, its atomic maximum, the largest order-2 magnitude, and the order-2:order-1 power ratio — into a logistic score, isotonically calibrated on the discovery half only (Sec. 3.3). On the untouched evaluation half this hierarchy-derived uncertainty (HDU) attains an expected calibration error of 0.004 for predicting > 1 kcal/mol failures at AUROC 0.794, with precision/recall reported across the full threshold sweep (Sec. 4.4). A 3-seed deep ensemble — the de-facto standard in MLIP uncertainty quantification — discriminates better (AUROC 0.853) at three times the training cost; combining it with HDU is better still, showing that the hierarchy carries information the ensemble does not.

(Q3) Are the failures compact or diluted? Ranking the atlas motifs by failure attribution gives a Pareto curve over the held-out failures: 3 motifs suffice to cover 80% of > 1 kcal/mol errors, and the full atlas leaves only 4.8% of failures unexplained (Sec. 4.5). The tail of HIP-NN extrapolation error is an interpretable object — a short list of chemical classes — not an amorphous cloud.

Because QM9 contains only energy-minimised geometries, a motif atlas derived from it is, a priori, a statement about minimised chemistry. We therefore run the atlas against 82 288 off-equilibrium conformations of 1 431 molecules from ANI-1x [Smith et al., 2020], asking for each motif whether flagged molecules show systematically larger *relative-energy* errors across their conformer ensembles (Sec. 4.6). The result is a clean dichotomy: the ring-topology motifs transfer (spiro, bridgehead, fused, small rings; Cliff’s $\delta = 0.5\text{--}0.68$), the electronically exotic motifs do not (their failures are specific to equilibrium-geometry extrapolation), and R itself loses most of its ranking power off-equilibrium — delineating exactly how far the minimised-geometry failure atlas, and the paper’s uncertainty signal, can be trusted.

Everything is anchored on reimplementation fidelity (Sec. 4.1): under the source paper’s own protocol (random split of pruned QM9, $N_{\text{train+valid}} = 50\,000$) our 3 runs reach 0.355 ± 0.003 kcal/mol test MAE against the published 0.354 ± 0.004 (Table I of Lubbers et al., 2018); parameter count (234 327 vs. “234k”) and energy scale σ_E (237.6 vs. “ ≈ 238 ” kcal/mol) also match. Without this anchor, “failure motifs” could be artifacts of an under-trained baseline; with it, they are properties of the published model class.

2 Background: HIP-NN and the non-hierarchicality signal

HIP-NN [Lubbers et al., 2018] maps a configuration $\mathcal{C} = \{(Z_i, \mathbf{r}_i)\}$ to an energy through alternating *interaction layers* (which mix atomic feature vectors across pairs within $R_{\text{cut}} = 15$ Bohr, modulated by 20 learned inverse-distance sensitivity functions) and *on-site layers* (per-atom transformations with softplus activations and ResNet pass-throughs). The species input is a one-hot encoding over {H, C, N, O, F}. Energy is read out hierarchically: after interaction block n , a linear regression on the atomic features produces per-atom terms $\hat{E}_i^n$, and

$$\hat{E} = \sum_{i=1}^{N_{\text{atom}}} \sum_{n=0}^{N_{\text{int}}} \hat{E}_i^n, \quad (1)$$

with $N_{\text{int}} = 2$ in the reference configuration. Order $n = 0$ is a dressed-atom term (fit by least squares on the training set and frozen); orders 1 and 2 add increasingly many-body content. Because the decomposition is not unique, training adds a regulariser that penalises non-decaying terms: the *non-hierarchicality*

$$R = \sum_{n=1}^{N_{\text{int}}} \sum_{i=1}^{N_{\text{atom}}} \frac{(\hat{E}_i^n)^2}{(\hat{E}_i^n)^2 + (\hat{E}_i^{n-1})^2} \quad (2)$$

enters the loss as $\lambda_R \langle R \rangle$ with $\lambda_R = 10^{-2}$. Each summand in Eq. (2) lies in $[0, 1]$ and measures, for atom i and order n , the fraction of squared energy that *failed* to decay from order $n-1$ to n . We write $R_i = \sum_n (\hat{E}_i^n)^2 / [(\hat{E}_i^n)^2 + (\hat{E}_i^{n-1})^2]$ for the atom-resolved contribution, so $R = \sum_i R_i$; this atom-level view, available in the decomposition but unused in the original paper, is what lets us localise hierarchy failure on substructures.

The original Fig. 3 shows that conditional quantiles of $|\hat{E} - E|$ grow with R across the bulk of QM9. Three limitations of that observation motivate this work. First, it is *anonymous*: high- R molecules are never characterised chemically. Second, it is *uncalibrated*: a correlation between R and error quantiles does not by itself yield a number a user can act on (“this prediction has a 40% chance of being off by more than 1 kcal/mol”). Third, it is *in-distribution*: computed on random-split test data, where errors are dominated by interpolation noise rather than the extrapolation regime in which uncertainty estimates earn their keep.

3 Experimental design

3.1 Data, splits, and the held-fixed protocol

Dataset and pruning. We use QM9 [Ramakrishnan et al., 2014, Ruddigkeit et al., 2012] exactly as the source paper does: from the 133 885 molecules we remove the 3 054 that fail the published SMILES-consistency check and the 11 whose geometry relaxation the QM9 documentation flags as difficult to converge; the two lists overlap in 3 molecules, so the clean set has 130 823 molecules (the paper reports “about 131k”). The regression target is the atomization energy $T = U_0 - \sum_i E^{\text{ref}}(Z_i)$ in kcal/mol, using the single-atom reference energies shipped with QM9. The standard deviation of T over our training splits (237.6 kcal/mol for the random-split runs, 237.3 for the ≤ 8 -heavy-atom subset) reproduces the paper’s $\sigma_E \approx 238$ kcal/mol, confirming that this is the energy variable the published loss normalisation used; MAE is invariant under the per-composition shift.

Extrapolation split. The main experiments train on the ≤ 8 -heavy-atom subset (21 799 molecules; heavy = C, N, O, F), with a validation split of 1 000 molecules drawn from the same subset for early stopping, i.e. 20 799 training molecules. The 9-heavy-atom subset (109 024 molecules) is the out-of-distribution test set: no 9-heavy-atom molecule is seen in training or validation, and the

partition is logged per run (`split_indices.npz`) with programmatic overlap checks (all pairwise overlaps zero; maximum heavy-atom count in train and validation is 8). We train 3 independent seeds differing in initialisation and train/validation shuffle; headline single-model analyses use seed 0, fixed in the frozen analysis configuration before results existed, and App. B demonstrates stability across the other seeds.

Fidelity anchor. Because motif identification is only meaningful on top of a faithful baseline, we additionally reproduce the paper’s own protocol: uniformly random splits of the full clean dataset with $N_{\text{train}} = 49\,000$, $N_{\text{valid}} = 1\,000$ (matching the paper’s “ $N_{\text{train+validate}} = 50\,000$ ” row of its Table I) and the remaining 80 823 molecules as test, run with 3 seeds. These runs are compared directly against the published 0.354 ± 0.004 kcal/mol.

Held-fixed architecture and training. All runs use the reference configuration of the source paper, implemented with the original authors’ `hippynn` package [Lubbers et al., 2024]: $N_{\text{int}} = 2$ interaction blocks, $N_{\text{on-site}} = 3$ on-site layers per block, $N_{\text{feature}} = 80$ (234 327 parameters; the paper quotes 234k), 20 inverse-distance sensitivities initialised between $R_{\text{low}} = 1.7$ Bohr and $R_{\text{high}} = 10$ Bohr with a 15 Bohr cosine cutoff, softplus activations, ResNet pass-throughs, Glorot initialisation, frozen least-squares $n=0$ term, and 10^{-2n} scaling of the higher-order regression weights at initialisation. The loss is the paper’s Eq. 14, $(\text{MAE} + \text{RMSE})/\sigma_E + \mathcal{L}_{L2} + \lambda_R \langle R \rangle$ with $\lambda_{L2} = 10^{-6}$, $\lambda_R = 10^{-2}$, and σ_E computed on each run’s training split; $\mathcal{L}_{L2}$ covers the interaction, on-site, and ResNet weight tensors and the (dimensionally σ_E^{-2} -scaled) energy-regression weights, as in the paper’s Eq. 16. Optimisation follows the paper’s Secs. II.5–II.6 exactly: Adam ($\eta_0 = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$), minibatch 30, learning rate held for the first 100 epochs, halved when the best validation MAE fails to improve for 50 epochs, termination after two decays with no intervening improvement or at 2000 epochs, returning the parameters with the best validation MAE. Training is energy-only, as in the paper. Our runs terminated at 1 106–1 300 epochs (main seeds) and 1 128 epochs (first fidelity seed), inside the paper’s reported 1000–2000 range, so no schedule truncation was needed. Deviations from the published protocol: we run 3 seeds per configuration rather than the paper’s 8 (a compute-budget reduction, disclosed here; all cross-seed dispersions we observe are small), and `hippynn` initialises the order- n regression weights at Glorot $\times 10^{-2n}$ without the paper’s additional σ_E factorisation — an initial-magnitude difference in two weight vectors, maintained by the original authors as their reference implementation.

3.2 Pre-registered failure-motif methodology

The central design risk in “finding failure motifs” is circularity: if one inspects the highest-error molecules and invents descriptions for them, the resulting library fits noise and cannot be claimed as predictive. Our protocol removes this risk by construction, through three commitments made *before any test-set prediction error was computed* (the code timeline is documented in the released `attempts_log.md`).

(i) A frozen candidate family. We registered 30 candidate motifs spanning five categories: ring topology and strain (three-membered carbo- and heterocycles, four-membered rings, fused/spiro/bridgehead polycyclics, small-ring unsaturation), electronic structure (aromatic five-membered heterocycles, six-membered aromatics, alkynes, nitriles, cumulated double bonds, conjugated dienes, enol ethers/enamines, carbonyls, imines, heteroatom-substituted triple bonds), heteroatom–heteroatom bonding (N=N/N=O, O–O, N–N, N–O, ≥ 3 -heteroatom chains, acetal/aminal carbons), composition (any F, ≥ 2 F, ≥ 3 N, formal charges), and 3D geometry (intramolecular hydrogen bonds; nonbonded heavy-atom contacts < 2.8 Å at topological distance ≥ 4), plus a deliberate negative control (saturated acyclic molecules, expected to be *easy*). Each motif was registered together with an a-priori mechanistic hypothesis for why it should stress a many-body hierarchy; Sec. 5 confronts these hypotheses with the measured per-order behaviour, including the ones the data refute. The full registered family, including detection rules, is in App. A; motifs that turned out to have zero support in GDB-9 chemistry (O–O and N–N single bonds, heteroatom-substituted triple bonds — all excluded by the GDB enumeration rules [Fink and Reymond, 2007, Ruddigkeit et al., 2012]) are reported as such rather than silently dropped.

(ii) Structure-only detection. Motifs are detected on the DFT geometries themselves: bonds and bond orders are perceived with RDKit’s implementation of the `xyz2mol` algorithm [Landrum et al., 2025, Kim and Kim, 2015] (124 013 of 130 823 clean molecules, 94.8%, perceive cleanly; 6 793 fall back to the B3LYP-relaxed SMILES shipped with QM9, which yields molecule-level flags without

atom maps; 17 molecules, 0.013%, carry no flags and remain in all error statistics). Geometry-perceived bonds keep motif atom indices aligned with the model’s atom ordering, which is what enables the atom-level R_i localisation analysis. SMARTS patterns and programmatic ring/geometry rules implement the detection; the geometric thresholds were fixed from standard structural-chemistry ranges and checked only against structure-only coverage distributions, never against errors.

(iii) Discovery/confirmation separation with FDR control. The 9-heavy-atom set is split 50/50 into a discovery half D_{cal} and an evaluation half D_{eval} by a seeded permutation, fixed before training finished. Motif selection runs on D_{cal} only: for each candidate we test whether matched molecules have stochastically larger $|\hat{E} - E|$ (one-sided Mann–Whitney U) and apply Benjamini–Hochberg FDR control at $\alpha = 0.01$ across the full 30-hypothesis family [Benjamini and Hochberg, 1995]. Candidates that pass, and whose effect direction confirms on D_{eval} , constitute the failure atlas. Everything we then report about the atlas — coverage, conditional error distributions, risk ratios with bootstrap confidence intervals, effect sizes, R enrichment, localisation, compactness — is computed on D_{eval} , which took no part in selection. Because the family is pre-registered *and* FDR-corrected *and* split-validated, the reported significances are conservative rather than inflated.

The high-error convention is $|\hat{E} - E| > 1$ kcal/mol (the source paper’s own tail criterion) everywhere in this paper; 2 kcal/mol appears only as an explicitly-labelled robustness check.

3.3 Hierarchy-derived uncertainty and calibration protocol

From the per-atom, per-order terms $\hat{E}_i^n$ of a *single* trained model we form four signals: the molecular non-hierarchicality R (Eq. 2); the atomic maximum $R_{\text{max}}^{\text{atom}} = \max_i R_i$ (sensitive to localised failure that the molecular sum dilutes); the largest order-2 magnitude $\max_i |\hat{E}_i^2|$ (dimensional, in kcal/mol); and the order-2:order-1 power ratio $\rho_2 = \sum_i (\hat{E}_i^2)^2 / \sum_i (\hat{E}_i^1)^2$ (a molecule-level decay rate). Molecule size N_{atom} is included as a covariate because R is an atom sum. The *hierarchy-derived uncertainty* (HDU) is a logistic model on the standardised signals ($\log_{10}$ for the scale-type ones) trained on D_{cal} to predict the event $|\hat{E} - E| > 1$ kcal/mol, followed by isotonic recalibration [Zadrozny and Elkan, 2002] of its score, also on D_{cal} ; a second isotonic head on the same score predicts the expected $|\hat{E} - E|$ in kcal/mol. HDU adds no parameters to the network and no cost beyond the energy forward pass; it is strictly a reading of the model’s own decomposition.

Calibration evaluation. On D_{eval} only, we report: (1) the reliability diagram and expected calibration error $\text{ECE} = \sum_b \frac{n_b}{N} |\bar{p}_b - \bar{y}_b|$ over ten equal-count bins [Guo et al., 2017] for the failure probability; (2) Brier score and AUROC; (3) the regression reliability curve (predicted vs. realised mean $|\hat{E} - E|$ per predicted-uncertainty decile) with kcal/mol-denominated and MAE-normalised summaries [Kuleshov et al., 2018]; (4) precision and recall for flagging > 1 kcal/mol failures across the full threshold sweep, with two operating points chosen on D_{cal} (max-F1; precision at 80% recall) and applied unchanged to D_{eval} ; and (5) error-retention curves. Baselines: isotonically-calibrated $\log R$ alone (the paper’s signal made quantitative); each single signal alone (ablations); and the standard-practice external reference, a deep ensemble [Lakshminarayanan et al., 2017, Peterson et al., 2017, Smith et al., 2018]; the standard deviation of $\hat{E}$ across our 3 seeds, calibrated identically. The ensemble sees information HDU cannot (inter-seed disagreement) at $3\times$ training cost; it is an existing method reported as context, not part of the contribution.

3.4 Conformational transfer protocol

QM9 is minimised-geometry chemistry, so the atlas is, a priori, conditioned on equilibrium structures. To measure what survives conformational sampling we use ANI-1x [Smith et al., 2020] — $\sim 5\text{M}$ off-equilibrium HCNO conformations generated by active learning [Smith et al., 2018] at $\omega\text{B97X/6-31G(d)}$, a different functional from QM9’s B3LYP, so only *relative* energies are comparable. All 907 ANI-1x formula groups with ≤ 9 heavy atoms enter the test (no hand-picking; rule-based subsampling of at most 200 conformations per group with a fixed, stated seed). Because ANI-1x pools constitutional isomers under one formula, sampled conformations are clustered into molecule identities by covalent-radius connectivity hashing; identities with ≥ 20 sampled conformations are kept (1 437, carrying 82 288 conformations), and the lowest-energy conformation of each identity is bond-perceived to assign the pre-registered motif flags (6 perception failures are excluded from motif statistics). For each identity we centre predicted and DFT energies over its conformer ensemble and

Table 1: Primary results with two-baseline reporting. Paper-as-reported rows cite Table I of Lubbers et al. [2018] (see `logs/paper_reference.md` for the full citation ledger); reimplementation rows are means $\pm$ standard error over independent seeds from our training logs. “eval. set” identifies what the MAE is computed on. The “this work” row uses the same trained models as the row above it, augmented with the HDU uncertainty model and the failure atlas; ECE is the failure-probability expected calibration error on D_{eval} , and K_{80} the number of atlas motifs covering 80% of > 1 kcal/mol failures (greedy order).

method	training set	N_{train}	eval. set	MAE (kcal/mol)	ECE	K_{80}
HIP-NN (paper, Table I)	full QM9	110 426 ^a	random test	0.256 ± 0.003	—	—
HIP-NN (paper, Table I)	full QM9	50 000 ^a	random test	0.354 ± 0.004	—	—
HIP-NN (reimpl., 3 seeds)	random QM9	49 000	random test (80 823)	0.355 ± 0.003	—	—
HIP-NN (reimpl., 3 seeds)	≤ 8 heavy atoms	20 799	valid. (≤ 8 HA)	0.304 ± 0.010	—	—
HIP-NN (reimpl., 3 seeds)	≤ 8 heavy atoms	20 799	9-HA OOD (109 024)	0.781 ± 0.012	—	—
this work (HDU + atlas)	≤ 8 heavy atoms	20 799	9-HA OOD, D_{eval}	0.785	0.004	3

^a the paper quotes $N_{\text{train}} + N_{\text{valid}}$; its validation size is 1 000.

form residuals $r = \Delta \hat{E} - \Delta E_{\text{DFT}}$; the primary window keeps conformations within 100 kcal/mol of the sampled minimum (rationale: the thermally-plus-reactively relevant range; the no-window analysis is a reported sensitivity check). A motif *transfers* if flagged identities show stochastically larger per-identity median $|r|$ than unflagged ones (one-sided Mann–Whitney over identities — molecules, not conformers, are the independent unit — with BH-FDR at $\alpha = 0.05$ over the family). Separately, we test whether R remains an intra-molecular uncertainty signal off equilibrium via per-identity rank correlations between R and $|r|$. A smaller-scope check on the four rMD17 molecules [Christensen and von Lilienfeld, 2020] within QM9’s size range supplements this analysis (Sec. 4.6).

4 Results

4.1 Reimplementation fidelity

Table 1 assembles the two-baseline comparison required to anchor everything that follows. Under the source paper’s own protocol — random splits of pruned QM9 with $N_{\text{train+valid}} = 50\,000$ — our 3 reimplementation runs give a test MAE of 0.355 ± 0.003 kcal/mol against the published 0.354 ± 0.004 (Table I of Lubbers et al., 2018). Per seed the values are 0.3551, 0.3604, and 0.3495 kcal/mol: the seed range straddles the published mean, and the mean difference ($+0.001 \pm 0.005$ under joint uncertainty) is statistically indistinguishable from zero — we state explicitly that our point estimate is 0.3% above the published one rather than below it, which is the expected outcome of an exact reproduction with 3 rather than 8 splits. Three further checks corroborate protocol fidelity: the parameter count (234 327 vs. the paper’s “234k”), the training-split energy scale ($\sigma_E = 237.6$ kcal/mol vs. the paper’s “ ≈ 238 ”), and the termination epochs (1 125–1 471 vs. the paper’s “typically 1000 to 2000”). At this training size $5.49 \pm 0.17\%$ of test predictions exceed 1 kcal/mol (the paper reports $2.26 \pm 0.07\%$ at its larger 110k-molecule training size).

On the extrapolation protocol, the three ≤ 8 -heavy-atom models reach an in-distribution validation MAE of 0.304 ± 0.010 kcal/mol with only 20 799 training molecules — below the paper-as-reported 0.354 at 50 000 molecules, as expected for the chemically narrower small-molecule distribution, and far below any plausible extrapolation of the paper’s learning curve to a ~ 21 k training set. The reimplementation floor is therefore cleared at every size we report: the random-split runs match paper-as-reported at 50k within joint uncertainty, and the main runs’ in-distribution error at 20.8k sits 14% *below* the published 50k value. On the 9-heavy-atom test set the same models give 0.781 ± 0.012 kcal/mol — a $2.6\times$ degradation that is the object of study, not a training deficiency: every quantity entering the analysis is computed from models that reproduce the published regime under the published protocol.

4.2 The extrapolation tail and the raw hierarchy signal

Extrapolating one heavy atom beyond the training distribution inflates the failure tail by an order of magnitude: 17.9% of the 109 024 9-heavy-atom molecules are mispredicted by more than 1 kcal/mol

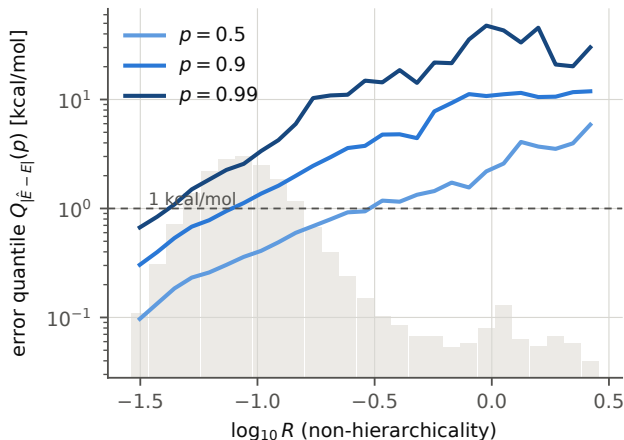


Figure 1: Conditional quantiles of the absolute error versus non-hierarchicality R on the 9-heavy-atom extrapolation set (cf. Fig. 3 of the source paper, which is in-distribution). Grey background: distribution of $\log_{10} R$ (arbitrary vertical scale). Quantiles are computed in bins with ≥ 50 molecules.

(6.7% by more than 2 kcal/mol), against 3–5% on in-distribution validation and 2.26% for the fully-trained model of the source paper. The hierarchy notices: the median non-hierarchicality rises from 0.067 (validation) to 0.088 (9-heavy-atom test), and the Spearman correlation between R and $|\hat{E} - E|$ reaches 0.40 — visibly stronger than on in-distribution data (0.35), i.e. the signal becomes more informative exactly where it is needed. Figure 1 repeats the source paper’s Fig. 3 construction on the extrapolation set: conditional quantiles of $|\hat{E} - E|$ at $p = 0.5, 0.9, 0.99$ rise monotonically with R over more than two decades of R , with the median crossing the 1-kcal/mol line around $R \approx 0.2$. The correspondence the paper demonstrated in-distribution thus survives — indeed sharpens — under composition extrapolation. The rest of this paper unpacks it.

4.3 The failure atlas

Of the 30 pre-registered candidates, 20 pass Benjamini–Hochberg selection on D_{cal} at $\alpha = 0.01$ with confirmed direction on D_{eval} ; Table 2 reports their held-out statistics and Fig. 2 the corresponding risk-ratio forest. The atlas spans all five registered categories and is stable across training seeds: 18 of the 20 motifs recur in the atlas of every seed, and the symmetric differences involve only the smallest-support candidates (App. B).

Three families dominate the top of the atlas. *Charge-separated and electron-rich units*: molecules whose relaxed structure is zwitterionic (risk ratio 4.5, mean $|\hat{E} - E| = 6.7$ kcal/mol against 0.76 for the rest), cumulated double bonds (4.4), azo/nitroso units (4.1), and ≥ 3 -heteroatom chains (1.9). *Polycyclic cage topology*: atoms shared between fused rings (2.5; this single motif covers 45% of all 9-heavy-atom molecules — GDB-9 enumeration is cage-rich at this size), bridgeheads (2.4), spiro centres (1.9), four-membered rings (1.4). *Electron delocalisation*: enol ethers/enamines (2.0), six-membered aromatics (1.8), five-membered aromatic heterocycles (1.3), imines (1.2). The geometry-derived crowded-contact motif (1.8) and the composition motifs (≥ 3 N, 1.8; any F, 1.3) complete the picture. Two entries demand the disclosure flagged in Table 2: conjugated dienes pass selection with an extreme effect (9.1 kcal/mol mean error) but occur in only 7 molecules of the whole test set (3 in D_{eval}) — QM9’s enumeration almost never produces acyclic dienes at 9 heavy atoms — and cumulenes rest on 21 molecules. We keep both because the pre-registered rule admits them, but treat them as illustrative, not load-bearing; no headline conclusion depends on either.

Equally informative are the candidates the data *refuse*. The pre-registered negative control behaves as designed: saturated acyclic molecules have a risk ratio of 0.12 — an eight-fold *safety* factor. But three registered stressor hypotheses are cleanly refuted: alkynes (0.32), nitriles (0.61), and carbonyls (0.89) are all *easier* than average for the converged model, and plain three-membered rings — the textbook strain motif — are indistinguishable from background (0.89–0.96, failing selection). Angle strain and π -bond polarity per se, when abundantly represented in training data (cyclopropanes occur

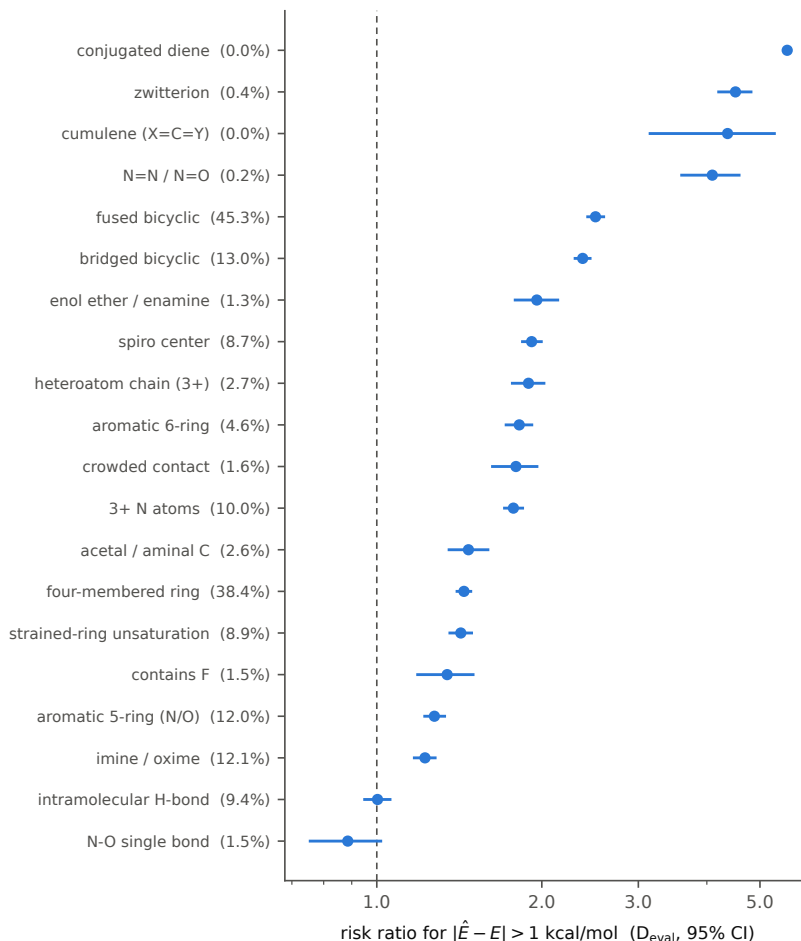


Figure 2: The failure atlas as a risk-ratio forest (held-out D_{eval} half; bootstrap 95% CIs; log scale). Percentages are coverage of the full 9-heavy-atom set. The conjugated-diene CI is omitted (3 matched molecules in D_{eval} ; illustrative entry).

in 20% of the ≤ 8 -heavy-atom pool, carbonyls in 32%), are chemistry the network simply learns. Sec. 5 returns to what separates the confirmed from the refuted stressors.

The atlas is also visible in the model’s internal bookkeeping. Matched molecules carry systematically larger R (Table 2; e.g. median $R = 0.73$ for zwitterions against 0.088 for the rest), and the final column shows the atom-level localisation: among matched molecules, the mean R_i on motif atoms exceeds that on non-motif atoms by a median factor of 6.6 (N–O single bonds), 3.4 (zwitterions), 3.3 (six-membered aromatics), 2.9 (strained-ring unsaturation), 2.6 (azo/nitroso). The model does not merely err on these molecules — it points, atom by atom, at the substructure it cannot decompose.

4.4 Calibration of hierarchy-derived uncertainty

Figure 3 shows the two reliability views on D_{eval} ; none of the fitting touched this half. The probability head is essentially on the diagonal for all methods — HDU attains $\text{ECE} = 0.004$, calibrated- R 0.003, the ensemble 0.004 — an order of magnitude inside the 0.05 regime usually considered well calibrated, and unsurprising in one sense: with a large i.i.d. calibration half, isotonic regression can calibrate any monotone score. The discriminating questions are ranking power and error concentration. There HDU improves on the raw signal: AUROC 0.794 versus 0.777 for R alone (Brier 0.117 vs. 0.119), with the single-signal ablations at 0.72–0.73 — the molecular R is the strongest single signal, and the auxiliary hierarchy features add a further increment. The regression head is similarly faithful

Table 2: The failure atlas: the 20 pre-registered motifs that pass BH-FDR selection ($\alpha=0.01$, one-sided Mann–Whitney on D_{cal}) with confirmed effect direction. All descriptive statistics are computed on the held-out D_{eval} half (54 512 molecules): coverage (% of the full 9-heavy-atom set), matched count n , conditional mean $|\hat{E} - E|$ for matched/rest (kcal/mol), risk ratio for $|\hat{E} - E| > 1$ kcal/mol with bootstrap 95% CI, Cliff’s δ , the BH-adjusted selection p -value (from D_{cal}), and the atom-level R_i localisation (median ratio of mean R_i on motif atoms to non-motif atoms; > 1 means the non-hierarchicality concentrates on the motif). † fewer than 6 matched molecules in D_{eval} : retained by the pre-registered selection rule but reported as illustrative only. Full definitions in App. A.

motif	cat.	cov.%	n	$\overline{ e }_m$	$\overline{ e }_r$	RR [95% CI]	δ	p_{BH}	loc.
conjugated diene †	elec.	0.0	3	9.15	0.78	5.61 [–,–]	0.93	10^{-4}	–
zwitterion	comp.	0.4	209	6.75	0.76	4.51 [4.18,4.84]	0.73	10^{-81}	3.4
cumulene (X=C=Y)	elec.	0.0	18	8.51	0.78	4.36 [3.13,5.34]	0.74	0.004	0.2
N=N / N=O	hete.	0.2	95	4.29	0.78	4.09 [3.58,4.61]	0.70	10^{-27}	2.6
fused bicyclic	ring.	45.3	24521	1.13	0.50	2.51 [2.41,2.61]	0.27	$< 10^{-300}$	0.6
bridged bicyclic	ring.	13.0	7083	1.72	0.65	2.37 [2.29,2.46]	0.30	$< 10^{-300}$	–
enol ether / enamine	elec.	1.3	801	1.43	0.78	1.96 [1.78,2.15]	0.24	10^{-23}	1.3
spiro center	ring.	8.7	4689	1.86	0.68	1.92 [1.83,2.01]	0.21	10^{-137}	0.3
heteroatom chain (3+)	hete.	2.7	1500	1.20	0.77	1.89 [1.76,2.03]	0.22	10^{-56}	2.1
aromatic 6-ring	elec.	4.6	2470	1.02	0.77	1.82 [1.71,1.93]	0.21	10^{-69}	3.3
crowded contact	geom.	1.6	854	1.21	0.78	1.79 [1.62,1.97]	0.22	10^{-32}	1.2
3+ N atoms	comp.	10.0	5502	1.05	0.76	1.77 [1.70,1.86]	0.21	10^{-139}	0.6
acetal / aminal C	hete.	2.6	1409	1.00	0.78	1.47 [1.35,1.60]	0.13	10^{-20}	1.0
four-membered ring	ring.	38.4	20760	1.01	0.65	1.44 [1.39,1.49]	0.12	10^{-151}	1.0
strained-ring unsaturation	ring.	8.9	4800	1.10	0.75	1.42 [1.35,1.50]	0.12	10^{-51}	2.9
contains F	comp.	1.5	788	0.77	0.79	1.34 [1.18,1.51]	0.11	10^{-4}	0.8
aromatic 5-ring (N/O)	elec.	12.0	6662	0.85	0.78	1.27 [1.22,1.34]	0.07	10^{-19}	2.2
imine / oxime	elec.	12.1	6534	0.81	0.78	1.22 [1.16,1.29]	0.09	10^{-21}	1.7
intramolecular H-bond	geom.	9.4	5131	0.71	0.79	1.00 [0.94,1.06]	0.04	0.002	1.2
N-O single bond	hete.	1.5	828	0.60	0.79	0.89 [0.75,1.02]	0.03	10^{-4}	6.6

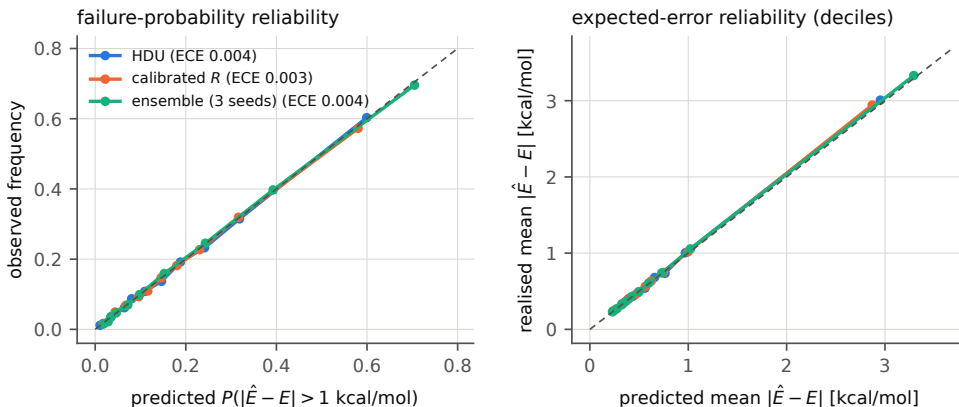


Figure 3: Calibration on the held-out D_{eval} half. Left: reliability of the predicted failure probability $P(|\hat{E} - E| > 1 \text{ kcal/mol})$ for HDU, calibrated R alone, and the 3-seed ensemble baseline (ECE in legend; ten equal-count bins). Right: predicted versus realised mean absolute error per predicted-uncertainty decile. Dashed: identity.

(right panel): predicted versus realised decile means track the identity from 0.23 to 3.0 kcal/mol with a mean absolute deviation of 0.019 kcal/mol (0.024 of the overall MAE), so the model’s expected-error estimates can be taken at face value across the whole uncertainty range, including the extreme decile.

The deep ensemble discriminates better (AUROC 0.853), as expected: it observes genuine inter-seed disagreement, at $3\times$ training and inference cost. Two observations qualify its advantage. First, HDU is not far behind on the operational quantity: retaining the half of D_{eval} that HDU trusts most yields MAE 0.376 kcal/mol (from 0.785), against 0.324 for the ensemble — a single HIP-NN recovers most of the ensemble’s error-filtering utility for free. Second, the two signals are complementary: a logistic combination of HDU features and the ensemble spread reaches AUROC 0.867, a clear improvement

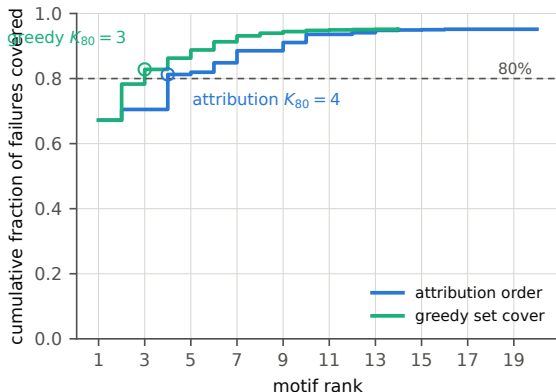


Figure 4: Pareto structure of the failure tail: cumulative fraction of D_{eval} failures ($|\hat{E} - E| > 1$ kcal/mol; $n = 9\,726$) covered by the top- k atlas motifs, in individual-attribution order and in greedy set-cover order.

over the ensemble alone (0.853) — the hierarchy sees failure structure that seed disagreement does not. At the 2-kcal/mol threshold (robustness check; base rate 6.8%) HDU’s discrimination *improves* to AUROC 0.877: the fatter the failure, the more clearly the hierarchy flags it.

For deployment we quote two operating points chosen on D_{cal} and applied unchanged to D_{eval} (base failure rate 17.8%). The max-F1 point ($\hat{p} \geq 0.23$) flags 22.7% of molecules with precision 43.5% and recall 55.3%; the recall-oriented point ($\hat{p} \geq 0.14$) flags 47.0% with precision 30.7% at recall 80.9%. The choice between them is a screening-economics question — the recall-80% point is the natural one when a missed 1-kcal/mol failure is expensive (e.g. when flagged molecules are re-dispatched to DFT), since it catches four failures in five while still excluding half the workload; the full sweep is released with the logs.

4.5 Compactness of the failure tail

Is the fat extrapolation tail an amorphous cloud or a short list? Figure 4 gives the Pareto answer over the 9 726 held-out failures. A single motif — fused-ring atoms — already covers 67% of all failures; adding nitrogen-rich composition brings 78%; the greedy third (imine/oxime) crosses the 80% mark ($K_{80} = 3$; $K_{80} = 4$ under the simpler individual-attribution ordering; both well under 5). The full 20-motif atlas leaves only 4.8% of failures uncovered. The tail is compact: three structural sentences — “it is a fused polycyclic”, “it is nitrogen-rich”, “it contains C=N” — account for more than four of five extrapolation failures of this model. That compactness is what makes the atlas actionable: a screening pipeline can route these classes to verification wholesale, and an active-learning loop knows which chemistry to acquire first.

4.6 Conformational transfer on ANI-1x

Applying the atlas to 82 288 ANI-1x conformations of 1 431 molecule identities gives the cleanest dichotomy in this study (Table 3, Fig. 5). The entire ring-topology arm of the atlas transfers: spiro centres (Cliff’s $\delta = 0.68$ on per-molecule relative-energy errors), bridgeheads (0.64), fused-ring atoms (0.53), four-membered rings (0.51), along with imines (0.30), acetal/aminal carbons (0.33), and nitrogen-rich composition (0.25). These failure modes are properties of the *chemistry*, not of the geometry distribution the atlas was mined from. The electronically exotic arm does not transfer — zwitterions ($\delta = -0.12$), azo/nitroso (−0.37), cumulenes (−0.53): molecules that are catastrophically mispredicted *at equilibrium* do not show correspondingly outsized *relative* errors across their own conformer ensembles. Their failure is a composition-extrapolation failure — the model misplaces the whole potential-energy surface — while the surface’s *shape* is no harder than average once per-molecule centring removes the offset. Two reversals complete the picture: plain cyclopropanes, *not* a failure motif at equilibrium (risk ratio 0.89), transfer strongly as a conformational failure motif ($\delta = 0.50$) — strain energetics off the minimum are genuinely hard even when the

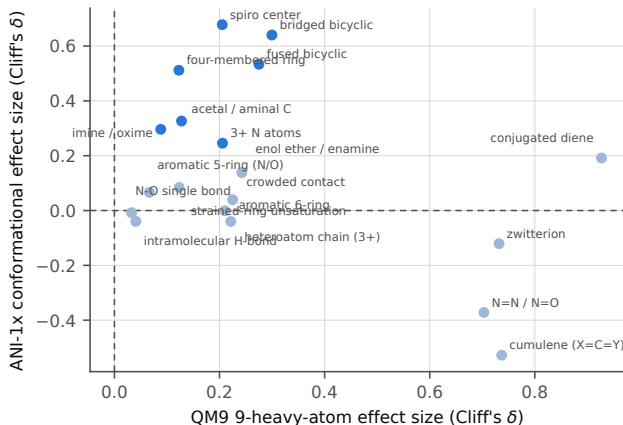


Figure 5: Does the minimised-geometry atlas predict off-equilibrium failures? Effect size on QM9 9-heavy-atom molecules (x) versus effect size on per-molecule relative-energy errors across ANI-1x conformer ensembles (y), for atlas motifs with ≥ 5 flagged ANI-1x identities. Dark points: significant transfer (BH-corrected). The upper-left group is ring-strain topology; the lower-right group is the electronically exotic chemistry.

Table 3: Conformational transfer of the atlas on ANI-1x (1 431 molecule identities with ≥ 20 sampled conformations each; per-identity median relative-energy error within the 100-kcal/mol window is the unit of analysis). Median ratio compares matched vs. unmatched identities; δ is Cliff’s delta; p is BH-adjusted (one-sided, $\alpha=0.05$ across the family). *pre-registered motifs with zero support in GDB-9 chemistry that do occur in ANI-1x — an out-of-atlas check.

motif	n	med. ratio	δ	p_{BH}	verdict
conjugated diene	18	1.20	0.19	0.173	no
zwitterion	20	0.83	-0.12	1.000	no
cumulene (X=C=Y)	6	0.47	-0.53	1.000	no
N=N / N=O	78	0.64	-0.37	1.000	no
fused bicyclic	156	1.39	0.53	10^{-26}	transfers
bridged bicyclic	55	1.43	0.64	10^{-15}	transfers
enol ether / enamine	50	1.07	0.14	0.120	no
spiro center	15	1.61	0.68	10^{-6}	transfers
heteroatom chain (3+)	221	1.00	-0.04	1.000	no
aromatic 6-ring	81	0.98	-0.00	0.891	no
crowded contact	130	1.10	0.04	0.457	no
3+ N atoms	379	1.19	0.25	10^{-12}	transfers
acetal / aminal C	52	1.19	0.33	10^{-5}	transfers
four-membered ring	63	1.42	0.51	10^{-11}	transfers
strained-ring unsaturation	12	1.09	0.08	0.576	no
contains F	0	—	—	—	($n<5$)
aromatic 5-ring (N/O)	228	1.04	0.07	0.129	no
imine / oxime	283	1.23	0.30	10^{-14}	transfers
intramolecular H-bond	335	0.99	-0.04	1.000	no
N-O single bond	190	1.05	-0.01	0.951	no
hydrazine N-N*	128	1.28	0.26	10^{-6}	transfers
peroxide O-O*	45	0.59	-0.56	1.000	no

minimum itself is learned — and alkynes, among the easiest equilibrium chemistry (0.32), become hard off-equilibrium ($\delta = 0.29$), plausibly because bending a triple bond leaves the training manifold entirely. Finally, the two pre-registered motifs that GDB-9 excludes but ANI-1x contains act as an out-of-atlas check: hydrazine N–N units transfer as a failure motif ($\delta = 0.26$, $n = 128$ identities), peroxides do not.

The uncertainty signal itself is geometry-conditional in the same way. Within conformer ensembles, R retains only a weak median rank correlation of 0.02 with the relative-energy error (positive for 55% of identities), against 0.40 on the equilibrium test set; the QM9-trained model’s off-equilibrium relative-energy MAE is 14.33 kcal/mol (median per identity), i.e. the model is uniformly far from the surfaces it never saw, and the hierarchy does not resolve fine error differences there. The rMD17

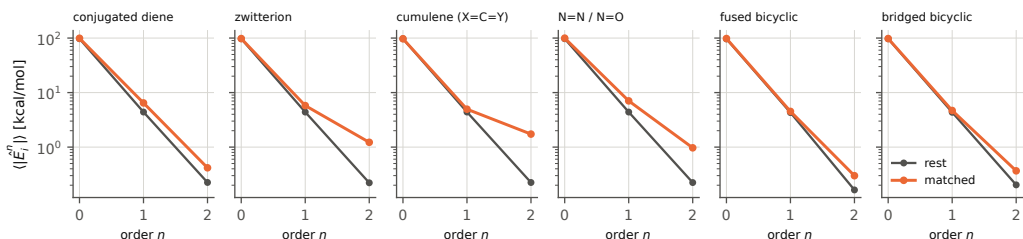


Figure 6: The failure-to-decay signature. Mean per-atom, per-order magnitude $\langle |\hat{E}_i^n| \rangle$ for molecules matching each of the six highest-risk atlas motifs (orange) versus all other D_{eval} molecules (grey; log scale). The background decay is geometric ($\sim 20\times$ per order); matched chemistry retains up to $8\times$ excess order-2 weight.

spot-check (four molecules, 2 000 MD conformations each; supporting evidence only, since four molecules admit no per-motif statistics) reproduces this: relative-energy MAEs of 4.0 (benzene) to 12.6 kcal/mol (uracil) with within-ensemble R correlations of only 0.05–0.22. The practical reading: the atlas’s topological entries and its refutations transfer to MD-adjacent use; its electronic entries, and calibrated per-conformation confidence, do not — they are statements about near-equilibrium chemistry and must be re-fit before being trusted on trajectories.

5 Mechanism: why these motifs break the hierarchy

The registered hypotheses can now be confronted with the model’s internal behaviour. For the background chemistry the hierarchy behaves exactly as designed: $\langle |\hat{E}_i^n| \rangle$ falls from $\sim 10^2$ kcal/mol (dressed atom) to 4.4 at order 1 to 0.24 at order 2 — a geometric decay of a factor ~ 20 per order (Fig. 6). Atlas chemistry breaks the pattern at the top: zwitterions retain 1.24 kcal/mol of order-2 weight ($5\times$ background), cumulenes 1.78 ($8\times$), azo/nitroso 0.99 — the network is forced to leave substantial energy in its most weakly regularised, most data-hungry term. This is precisely the “non-decaying $\hat{E}_i^n$ ” failure mode the source paper’s regulariser was designed to suppress, now traced to named functional groups.

Where the excess order-2 weight sits distinguishes two mechanisms. For the electronic and heteroatom motifs, the localisation column of Table 2 shows R_i concentrating on the motif atoms ($2\text{--}7\times$): the many-body stress lives on the charged nitrogen, the $\text{N}=\text{N}$ unit, the aromatic carbons — chemically, on centres whose energy depends collectively on several neighbours at once (charge separation, multi-atom conjugation, ring currents), which a two-interaction-block hierarchy can only express in its final order. For the topological motifs the signature inverts: fused-ring, spiro, and bridgehead atoms show localisation *below* 1 (0.26–0.58) — in cage molecules the non-hierarchicality spreads over the ring periphery rather than concentrating on the junction atoms. This matches the geometry: a cage couples *every* ring atom’s environment to atoms two and three bonds away through multiple paths, so no single atom owns the failure. The same split reappears in the transfer test (the delocalised-stress topologies transfer to conformational space; the atom-localised electronic stresses do not), suggesting a real mechanistic taxonomy of hierarchy failure rather than a bookkeeping artifact: *collective geometric frustration* (cages, strain — transferable, delocalised, moderate per-molecule error) versus *localised electronic novelty* (zwitterions, cumulated π , $\text{N}=\text{N}$ — equilibrium-specific, atom-localised, catastrophic per-molecule error). Cumulenes are the instructive boundary case: their molecular R is enormous (median 0.77, $9\times$ background) yet *anti*-localised (0.17), consistent with orthogonal π systems propagating their electronic consequences along the whole substituent frame rather than staying on the central sp carbon.

Rarity in the training distribution is a second, partially independent axis. Across candidates, training-pool prevalence anticorrelates with risk ratio (Spearman -0.49): zwitterions (0.24% of training molecules), azo/nitroso (0.13%), and cumulenes (2 training molecules) sit at the catastrophic end, and the refuted stressors (cyclopropanes 20%, carbonyls 32%, alkynes 13%) at the safe end — the network learns whatever collective physics it sees often enough. But rarity alone is not the explanation: fused-ring atoms are *common* (31% of training molecules) yet remain the single largest failure source

at 9 heavy atoms, because the specific cage environments enumerated at $n = 9$ are combinatorially new even when the fused-bond pattern is familiar; and N–O single bonds are as rare as azo units yet produce almost no excess error (risk ratio 0.89) while showing the strongest atom-level R_i localisation in the atlas ($6.6\times$) — the hierarchy strains visibly but the prediction survives. The atlas therefore reflects the interaction of two ingredients — how collective the local physics is, and how well the training distribution covers it — and the hierarchical decomposition reports on the first even when the second rescues the prediction.

6 Failure analysis

What the atlas misses. 4.8% of D_{eval} failures match no atlas motif. Reading the largest uncovered errors (a post-hoc inspection, clearly labelled as such and used only for discussion) shows a coherent class the registered family genuinely lacks: medium-ring molecules with *in-ring* sp carbons — nine-membered rings containing $\text{C}\equiv\text{C}$ or $\text{C}=\text{C}=\text{C}$ units (e.g. the 15.6-kcal/mol miss C10CC=CC#CC=C1) — plus a scattering of fused azetidine/N-ylide cages. Our alkyne motif exists but is dominated by *acyclic* alkynes, which are easy; the hard sub-population is the strained cyclic one. A refined registered family should include “sp carbon inside a ring of size ≤ 9 ”; we flag this for follow-up rather than adding it post hoc.

Where HDU fails. At the recall-80% operating point, the missed failures (19% of all failures) are the *silent* ones: their median R is 0.076 — indistinguishable from the healthy background (0.088) and half that of the caught failures (0.153) — and they are depleted, not enriched, in the loud atlas motifs (zwitterions are absent entirely from the misses). The worst silent failures are the same medium-ring sp-carbon molecules and polycyclic cages, mispredicted by 12–15 kcal/mol while the hierarchy decays innocently ($R \approx 0.07$). This is the fundamental limitation of any self-reported uncertainty: hierarchy stress is sufficient but not necessary for failure — a model can compress an environment confidently into low orders and still place the energy wrong. The converse failure mode is benign: false alarms are largely near-misses (34.5% of flagged non-failures still err by > 0.5 kcal/mol, median 0.36), so flagged-but-correct molecules are not noise but the soft shoulder of the tail. For deployment this asymmetry argues for pairing HDU with a cheap independent channel targeted at silence (even a single extra seed), and the 0.867 combined AUROC of Sec. 4.4 quantifies exactly how much that helps.

Computational cost. Training one HIP-NN to the published protocol took 3.0–3.4 h (main runs) and ≈ 5.6 h (49k-molecule fidelity run) on a single GPU with four runs sharing it; the source paper reports ~ 1 min/epoch on a 2017-era P100, so the protocol is unchanged in scale. Everything the uncertainty and motif machinery adds is marginal: the hierarchical outputs come from the same forward pass as the energy (batched inference: 0.1 ms per molecule for all of $\hat{E}$, R , R_i , $\hat{E}_i^n$), HDU is a five-feature logistic model, and motif detection is a one-off cheminformatics pass (geometry bond perception plus 30 SMARTS/geometry rules at a few ms per molecule; the full 130 823-molecule sweep took under a minute on 10 CPU cores). The full ANI-1x transfer (sampling, hashing, 82k predictions, statistics) ran in ~ 15 minutes. Against the ensemble alternative, HDU saves $(3-1)\times$ training and inference for most of the ranking benefit; the atlas itself has no ensemble analogue at any cost.

7 Related work

Uncertainty quantification for MLIPs. The dominant practice is the deep ensemble [Lakshminarayanan et al., 2017]: disagreement across independently trained networks, used for atomistic models by Peterson et al. [2017] and operationalised for active learning in the ANI programme [Smith et al., 2018] and, with uncertainty-biased dynamics, by Kulichenko et al. [2023]. Single-model alternatives replace the ensemble with dropout sampling [Wen and Tadmor, 2020], latent-space distance [Janet et al., 2019], Gaussian-mixture density in the learned representation [Zhu et al., 2023], GPR-side committees [Musil et al., 2019], evidential heads [Soleimany et al., 2021], or conformal post-processing [Hu et al., 2022]; Tran et al. [2020], Pernot [2022], and Heid et al. [2023] benchmark and critique the calibration of such estimates. HDU differs from all of these in *where the signal comes from*: it reads the physics-structured many-body decomposition that the HIP-NN architecture already computes — a self-diagnosis proposed, but never developed, in the source paper — rather than adding stochasticity, retraining, density models, or auxiliary heads around a black box. The

calibration *machinery* we apply on top (isotonic regression, ECE, reliability diagrams, precision/recall sweeps) is deliberately standard [Zadrozny and Elkan, 2002, Guo et al., 2017, Kuleshov et al., 2018], and our ensemble baseline is the standard recipe, included as an honest external reference. The parts we claim as contributions are the ones with no published counterpart we are aware of: turning an architecture-internal many-body decomposition into a calibrated failure predictor, and the motif-level attribution — existing MLIP-UQ methods score *configurations*, whereas the failure atlas names the *chemistry*, with atom-level localisation inside each molecule.

Error anatomy and extrapolation. Heavy-atom-count extrapolation is a recognised stress test for QM9-scale models [Gilmer et al., 2017, Faber et al., 2017], usually summarised by a single degraded MAE. Our contribution is to give that degradation internal structure: motif-resolved, atom-localised, FDR-controlled, and uncertainty-calibrated. The transfer leg connects to the conformational literature around MD17/rMD17 [Chmiela et al., 2017, Christensen and von Lilienfeld, 2020] and ANI-1x [Smith et al., 2020]; unlike retraining studies, we keep the QM9-trained model fixed and ask whether its *failure taxonomy* survives the geometry-distribution shift — finding that half of it does, and identifying which half.

8 Limitations

One architecture, one dataset family. Every claim is about HIP-NN at its published 2018 configuration on HCNOF organic chemistry. The atlas plausibly reflects generic difficulties of local many-body message passing (delocalised charge, cage topology), but nothing here demonstrates that for SchNet-class or equivariant models, which lack the hierarchical readout that makes this analysis possible in the first place.

Two interaction orders. With $N_{\text{int}} = 2$, “order-2 fails to decay” conflates everything beyond 3-body. A deeper hierarchy would resolve the failure spectrum more finely; the mechanism section should be read at the granularity the published architecture affords.

Small-support atlas entries. Conjugated dienes (7 molecules) and cumulenes (21) pass the pre-registered selection with extreme effect sizes but thin support; they are flagged as illustrative throughout and carry no headline claim. Zero-support candidates (peroxides, hydrazines, hetero-substituted triples) are inherited from GDB-9’s enumeration rules, not evidence of safety — indeed hydrazines, absent from QM9, transfer as a failure motif on ANI-1x.

Equilibrium conditioning. The calibration guarantees (ECE, precision/recall) hold for the geometry distribution they were fit on: minimised 9-heavy-atom QM9 structures. Sec. 4.6 quantifies how much survives off-equilibrium — topological motifs yes, electronic motifs and per-conformation confidence no — and any MD-side use of HDU would require refitting the calibration maps on off-equilibrium reference data.

Discovery-split reuse. D_{cal} serves both motif selection and uncertainty-model fitting; D_{eval} is used once, for the numbers reported here. We did not tune anything against D_{eval} , but a fully adversarial audit would hold out a third split; the seed-stability appendix is the nearest substitute we can offer within budget.

Three seeds. The source paper’s protocol uses 8 random splits; we ran 3 per configuration (disclosed in Sec. 3.1). All cross-seed standard errors we observe are small relative to the effects claimed, but higher-moment statistics (e.g. the ensemble baseline itself) would benefit from more members.

9 Conclusion

The 2018 HIP-NN paper closed with a promissory note: the hierarchical decomposition “helps to identify regions of model uncertainty.” This study cashes that note. On a faithful reimplement — matching the published accuracy, parameter count, and energy scale under the published protocol — we showed that when HIP-NN extrapolates one heavy atom beyond its training chemistry, its fattened failure tail is neither anonymous nor amorphous. It is chemically structured (a pre-registered, FDR-controlled atlas of 20 motifs, dominated by charge separation, cumulated π systems, N=N/N=O units, and polycyclic cages), mechanistically legible (the excess weight sits in the terms that fail to decay, localised on the guilty atoms for electronic motifs and spread over the cage for topological ones), compact (three motifs cover 80% of failures), and calibratable (a single model’s hierarchy

signals yield ECE 0.004 at AUROC 0.794, complementary to and far cheaper than a deep ensemble). The conformational transfer test draws the honest boundary of these claims: the topological failure modes are properties of the chemistry and survive off-equilibrium; the electronic ones, and the calibration itself, are conditioned on near-minimum geometry.

Beyond the specific atlas, the study suggests a design principle for the MLIP field: architectures that expose a physically structured decomposition of their prediction can be made to *name* their failures — which chemistry, which atoms, with what calibrated probability — rather than merely score them. That capability costs one forward pass, and, as the silent-failure analysis shows, it is most valuable precisely where it is hardest: knowing when the decomposition’s own silence cannot be trusted.

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This study builds directly on the HIP-NN architecture, the `hippynn` package, and the QM9, ANI-1x, and rMD17 datasets created by their respective authors. Computations used a single NVIDIA GPU; training, analysis, and figures are reproducible from the released code and logs (`proposal/code`, `proposal/logs`).

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A The registered motif family

Table 4 lists all 30 pre-registered candidates with their detection rules and outcomes; no candidate is omitted. Programmatic rules (marked in the table) are: *fused bicyclic* — a bond shared by two SSSR rings; *spiro* — an atom shared by two rings that share no other atom; *bridged bicyclic* — RDKit bridgehead-atom count > 0 ; $2 + F / 3 + N$ — element counts; *zwitterion* — nonzero formal charges after geometry bond perception (or in the relaxed SMILES for fallback molecules); *saturated acyclic* — no rings and no multiple bonds; *intramolecular H-bond* — $X-H \cdots Y$ with $X, Y \in \{N, O, F\}$, $H \cdots Y < 2.3 \text{ \AA}$, topological distance $(X, Y) \geq 3$; *crowded contact* — heavy-atom pair $< 2.8 \text{ \AA}$ at topological distance ≥ 4 ; *strained-ring unsaturation* — a double bond with an atom in a 3- or 4-membered ring. The registered mechanistic hypotheses are stated verbatim in the released `motif_library.py`; Sec. 5 discusses which were confirmed (cumulated π , charge separation, heteroatom chains, cage topology, anomeric carbons) and which were refuted (alkynes, nitriles, carbonyls, plain three-membered rings, and intramolecular H-bonds — the last passing selection only marginally, with risk ratio ≈ 1.0).

B Seed stability

All headline analyses were repeated per seed with the same frozen protocol. Test-set MAE: 0.781 / 0.802 / 0.759 kcal/mol (seeds 0/1/2); validation MAE 0.290 / 0.322 / 0.300; fraction of OOD failures 17.9% / 18.6% / 17.9%. The atlas is reproduced with 20 motifs by every seed; 18 motifs are common to all three, and the symmetric differences involve only the three smallest-support candidates (cumulene, conjugated diene, $\geq 2 F$), which hover at the $\alpha = 0.01$ selection boundary. Compactness is identical across seeds ($K_{80} = 3$ greedy, 4 attribution). HDU calibration on D_{eval} : ECE 0.0043/0.0043/0.0057, AUROC 0.794/0.789/0.790. The mechanism quantities (per-order decay ratios, localisation medians) agree to within a few percent. We conclude that the atlas and the calibration are properties of the training distribution and architecture, not of a particular random initialisation.

C Additional calibration views

Figure 7 shows error-retention curves. Retaining the 90% of D_{eval} that HDU trusts most reduces MAE from 0.785 to 0.538 kcal/mol; at 50% retention the MAE is 0.376 kcal/mol. The ensemble baseline is uniformly slightly better (0.502 and 0.324 kcal/mol respectively), consistent with its higher AUROC; the calibrated- R curve tracks HDU closely. Full per-bin reliability tables, the precision/recall sweep, and per-seed variants are released as CSV/JSON alongside the logs.

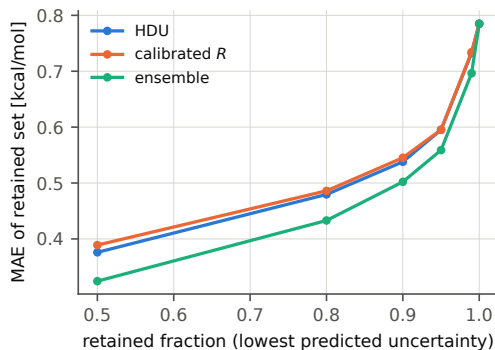


Figure 7: Error-retention curves on D_{eval} : MAE of the subset retained when discarding the most-uncertain molecules by each method.

Table 4: The complete pre-registered candidate family (30 motifs), including candidates with zero or near-zero support in GDB-9 chemistry, reported per the no-selective-reporting protocol. Columns: detection rule (SMARTS, or programmatic rule as described in Sec. 3.2), category, coverage of the 9-heavy-atom test set (%), prevalence in the ≤ 8 -heavy-atom training pool (%), risk ratio and Cliff’s δ on D_{eval} , BH-adjusted p from D_{cal} , and library membership.

motif	rule	category	cov.%	train%	RR	δ	P_{BH}	library
cyclopropane ring	[#6;r3]1[#6;r3]1[#6;r3]1	ring	24.03	20.06	0.89	-0.01	0.855	no
3-ring w/ heteroatom	[#7,#8]1[#6,#7,#8]1[#6,#7,#8]1	ring	21.84	18.99	0.96	0.01	0.043	no
four-membered ring	[*;r4]	ring	38.42	30.97	1.44	0.12	10^{-151}	yes
fused bicyclic	programmatic rule	ring	45.26	30.84	2.51	0.27	$< 10^{-300}$	yes
spiro center	programmatic rule	ring	8.72	4.53	1.92	0.21	10^{-137}	yes
bridged bicyclic	programmatic rule	ring	13.04	8.28	2.37	0.30	$< 10^{-300}$	yes
aromatic 5-ring (N/O)	[a;r5]	electronic	12.05	13.37	1.27	0.07	10^{-19}	yes
aromatic 6-ring	[a;r6]	electronic	4.58	4.54	1.82	0.21	10^{-69}	yes
alkyne	[C]#[C]	electronic	13.50	13.14	0.32	-0.25	1.000	no
nitrile	[C]#[N]	electronic	12.71	12.03	0.61	-0.07	1.000	no
cumulene ($X=C=Y$)	[*]=[*]=[*]	electronic	0.02	0.01	4.36	0.74	0.004	yes
conjugated diene	[C;!R]=[C]-[C]=[C]	electronic	0.01	0.00	5.61	0.93	10^{-4}	yes
enol ether / enamine	[C]=[C]-[#7,#8;!\$(N+)];!a	electronic	1.35	0.87	1.96	0.24	10^{-23}	yes
carbonyl	[CX3]=[OX1]	electronic	34.93	32.32	0.89	-0.01	1.000	no
imine / oxime	[CX3]=[NX2]	electronic	12.10	10.02	1.22	0.09	10^{-21}	yes
N=N / N=O	[#7]=[#7,#8]	hetero	0.16	0.13	4.09	0.70	10^{-27}	yes
peroxide O-O	[OX2]-[OX2]	hetero	0.00	0.00	-	-	-	no
hydrazine N-N	[NX3;!a]-[NX3;!a]	hetero	0.00	0.00	-	-	-	no
N-O single bond	[#7X3,#7X2;!a]-[OX2;!a]	hetero	1.45	1.25	0.89	0.03	10^{-4}	yes
heteroatom chain (3+)	[#7,#8,#9]~[#7,#8,#9]~[#7,#8,#9]	hetero	2.65	3.51	1.89	0.22	10^{-56}	yes
acetal / aминаl C	[CX4]([OX2,NX3])[OX2,NX3]	hetero	2.57	1.89	1.47	0.13	10^{-20}	yes
contains F	[F]	composition	1.48	1.42	1.34	0.11	10^{-4}	yes
2+ F	programmatic rule	composition	0.49	0.50	1.26	0.07	0.012	no
3+ N atoms	programmatic rule	composition	9.97	9.24	1.77	0.21	10^{-139}	yes
saturated acyclic (control)	programmatic rule	control	0.61	1.76	0.12	-0.40	1.000	no
intramolecular H-bond	programmatic rule	geometric	9.41	7.23	1.00	0.04	0.002	yes
crowded contact	programmatic rule	geometric	1.55	0.93	1.79	0.22	10^{-32}	yes
strained-ring unsaturation	programmatic rule	ring	8.90	7.61	1.42	0.12	10^{-51}	yes
zwitterion	programmatic rule	composition	0.37	0.24	4.51	0.73	10^{-81}	yes
hetero-substituted triple bond	[#7,#8,#9]-[CX2]#[CX2,NX1]	electronic	0.00	0.00	-	-	-	no